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Class: III - AIDS - A
Subject: Exploratory data Analysis
Date: 01/08/2026
Assignment - I

①

Department	Employee	Salary
IT	A	50000
IT	B	60000
HR	C	45000
HR	D	55000
IT	E	70000
HR	F	65000

To display dataframe:

```
import pandas as pd
data = {
    "Department":["IT","IT","HR","HR","IT","HR"],
    "Employee": ["A","B","C","D","E","F"],
    "Salary": [50000,60000,45000,55000,70000,65000]
}
df = pd.Dataframe(data)
print ("Employee Data")
print(df)
```

② Pivot table to calculate average salary for eac

```
import pandas as pd
data = {
        "Department":
        "Employee":
        "Salary":
       }
df = pd.DataFrame(data)
pivot_avg = pd.pivot_table(
                          df,
                          values = "Salary",
                          index = "Department",
                          aggfunc = "mean"
                         )

print("Average salary By Department")
print(pivot_avg)
```

Output:

```
Department
HR        55000
IT        60000
```

pivot table to calculate tot. salary for each dep (sum)

```
import pandas as pd
   :
   :

df = pd.DataFrame(data)
pivot_sum = pd.pivot_table(
                    df,
                    values = "Salary",
                    index = "Department",
                    aggfunc = "sum"
                    )

print("Total Salary by department")
print(df)
```

output:

```
Department:
HR    165000
IT    180000
```

④ Count no. of Employees' in each department:

```
pivot_count = pd.pivot_table(
                    df,
                    values = "Employee"
                    index = "Department"
                    aggfunc = "count"
                    )

print("Employee Count")
print(df)
```

output:

Department

HR 3

IT 3

⑤ max, min salary for each department:

```
Pivot_max_min = pd.pivot_table(
                    df,
                    values = "salary",
                    index = "Department",
                    aggfunc = ["max", "min"]
                    )

print(pivot_max_min)
```

output:

Department	max	min
HR	65000	45000
IT	70000	50000

⑥ multiple aggregation functions:

```
Pivot-all = pd.pivot_table(
                df,
                values = "salary",
                index = "department",
                aggfunc = ["mean", "sum", "count",
                           "max", "min"]
                )

print (pivot_all)
```

scenario question 1

① customer age vs purchase amount

```
import pandas as pd
data = {
    "customer": ["C1", "C2", "C3", "C4", "C5"],
    "Age": [20,20,25,29,30]
    "purchase Amount": [1500,2000,3000,2500,3500]
    }

df = pd.DataFrame(data)
pivot = pd.pivot_table(
                    df,
                    values = "purchase Amount",
                    index = "Age",
                    aggfunc= "mean"
                    )

print (pivot)
```

scenario question 2

① sample dataframe

```
import pandas as pd
data = {
    "Department": ["IT", "IT", "HR", "HR", "Sales", "IT", "IT",
                   "HR", "Sales", "Finance"],
    "Gender": ["M", "F", "M", "M", "M", "F", "M", "F", "F",
               "F"],
    "Salary": [45000, 50000, 55000, 43000, 57000, 60000,
               65000, 53000, 90000, 98000]
    }
```

```
df = pd.DataFrame(data)
print(df)
```

ii) Avg. Salary:

```
avg_salary = pd.pivot_table(
                    df,
                    values = "salary",
                    index = "Department",
                    aggfunc = "mean"
                    )

print(avg_salary)
```

iii) Tot. salary by gender:

```
gender_salary = pd.pivot_table(
                    df,
                    values = "salary",
                    index = "gender",
                    aggfunc = "sum"
                    )

print(gender_salary)
```

iv) max, min salary

```
max_min = pd.pivot_table(
                    df,
                    value = "salary",
                    index = "department",
                    aggfunc = ["max", "min"]
                    )

print(max_min)
```

Employee count by department:

```
count = pd.pivot_table(
                        df,
                        values= "salary",
                        index = "department",
                        aggfunc= "count"
                      )

print(count)
```